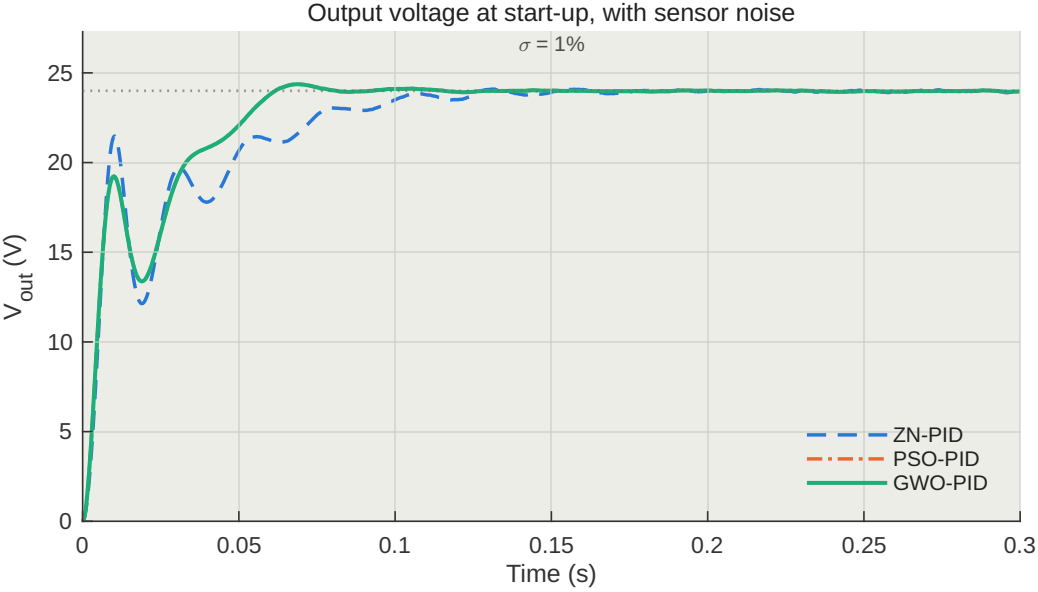

Output voltage at start-up, with sensor noise
σ = 1%
V_out (V)
Time (s)
ZN-PID
PSO-PID
GWO-PID
0
5
10
15
20
25
0
0.05
0.1
0.15
0.2
0.25
0.3